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I. DIFFUSION MODELS

Diffusion models [1, 2] are a class of generative models that learn a target distribution $p_{\rm data}(\phi)$ by inverting a gradual noising process. In the forward direction, a data sample $\phi_0 \sim p_{\rm data}$ is progressively corrupted into pure noise $\phi_{t_{\rm max}}$ over a fictitious time interval $t \in [0, t_{\rm max}]$. In the reverse direction, the model learns to denoise, thereby transporting samples from an analytically tractable prior back to the data distribution. Unlike one-shot generators such as GANs or VAEs, diffusion models decompose the generative task into a sequence of local denoising problems, each individually tractable and composing into an expressive and stable sampler. This structure is especially well suited to lattice field configurations. High-quality training data can be produced by standard Monte Carlo, and because the action $S[\phi]$ is known in closed form, the generated samples can be exactness-corrected against $e^{-S[\phi]}$ via a Metropolis-adjusted Langevin (MALA) step driven by the learned score.

A. Forward process as a stochastic differential equation

The forward noising process can be formulated as an Itô stochastic differential equation [2]

$$\mathrm{d}\phi \;=\; f(\phi,t)\,\mathrm{d}t \;+\; g(t)\,\mathrm{d}w, \tag{1}$$

where $f(\phi,t)$ is the drift term, $g(t)$ the diffusion coefficient, and w is a Wiener process with independent components at every spatial point. Equivalently, the Langevin form reads

$$\begin{aligned} \frac{\partial \phi_t(x)}{\partial t} \;&=\; f\big(\phi_t(x),t\big) \;+\; g(t)\,\eta(t,x), \\ \big\langle \eta(t,x)\,\eta(t',x')\big\rangle \;&=\; \delta(x-x')\,\delta(t-t'), \end{aligned} \tag{2}$$

with η a Gaussian white noise uncorrelated between distinct spatial points x. Two canonical choices of (f,g) have emerged.

a. Variance-preserving(VP) The variance-preserving choice takes $f(\phi,t) = -\frac{1}{2}\beta(t)\phi$ and $g(t) = \sqrt{\beta(t)}$, with $\beta(t) > 0$ controlling both the drift and the noise strength. The drift is proportional to ϕ with a negative coefficient, and hence acts as a restoring force towards zero. Integrating Eq. (1) over the interval $[0,t]$ from the initial condition ϕ_0, the distribution of ϕ_t conditioned on ϕ_0 is a Gaussian,

$$p_{0t}\big(\phi_t \,\big|\, \phi_0\big) \;=\; \mathcal{N}\Big(\phi_t;\, \sqrt{\bar\alpha(t)}\,\phi_0,\, (1-\bar\alpha(t))\,\mathbb{1}\Big), \tag{3}$$

with $\bar\alpha(t) = \exp\big(-\int_0^t \beta(s)\,\mathrm{d}s\big)$ decreasing from 1 at $t=0$ to 0 as $t\to\infty$. The conditional mean $\sqrt{\bar\alpha(t)}\,\phi_0$ shrinks from ϕ_0 towards zero while the conditional variance $1-\bar\alpha(t)$ grows from 0 to 1, so the endpoint distribution approaches the standard Gaussian $\mathcal{N}(0,\mathbb{1})$ regardless of ϕ_0. For unit-variance data, $\mathrm{Var}[\phi_t] = \bar\alpha(t)\,\mathrm{Var}[\phi_0] + (1-\bar\alpha(t)) = 1$ at all t, giving the schedule its name *Variance-preserving*. Eq. (3) is also the continuous-time limit of the DDPM formulation of Ho *et al.* [1].

b. Variance-exploding(VE). The variance-exploding choice takes $f(\phi,t) = 0$ and a positive diffusion coefficient $g(t)$ as a prescribed noise schedule, so the SDE reads

$$\mathrm{d}\phi \;=\; g(t)\,\mathrm{d}w. \tag{4}$$

The absence of drift leaves the initial field ϕ_0 untouched while noise of growing variance is injected on top, from which the schedule takes its name. Integrating Eq. (4) over the interval $[0,t]$ from the initial condition ϕ_0, the distribution of ϕ_t conditioned on ϕ_0 is a Gaussian,

$$p_{0t}\big(\phi_t \,\big|\, \phi_0\big) \;=\; \mathcal{N}\big(\phi_t;\, \phi_0,\, \sigma^2(t)\,\mathbb{1}\big), \tag{5}$$

with conditional mean fixed at ϕ_0 and conditional variance $\sigma^2(t)\,\mathbb{1}$, where $\sigma^2(t) = \int_0^t g(s)^2\,\mathrm{d}s$ is the accumulated noise variance and grows monotonically from 0 at $t=0$. The terminal noise scale $\sigma_{\rm max}$ is typically chosen following the prescription of Song & Ermon [3], $\sigma_{\rm max} \gtrsim \max_{i,j}\|\phi_i - \phi_j\|_2$, where ϕ_i, ϕ_j run over pairs of training configurations. This bound ensures that the noise-smeared distribution $p_{\sigma_{\rm max}}(\phi) = \int p_{\rm data}(\phi')\,\mathcal{N}\big(\phi;\phi',\sigma^2_{\rm max}\mathbb{1}\big)\,\mathrm{d}\phi'$ is effectively a single Gaussian that retains essentially no information about the initial field ϕ_0. This formulation corresponds to the NCSN/SMLD line of work [4].

Throughout this work we adopt the VE form with the geometric schedule $g(t) = \sigma^t$ on $t\in[0,1]$, parametrised by a single scalar hyperparameter $\sigma > 1$. For this choice the accumulated variance evaluates in closed form,

$$\sigma^2(t) \;=\; \int_0^t g(s)^2\,\mathrm{d}s \;=\; \frac{\sigma^{2t}-1}{2\ln\sigma}, \tag{6}$$

with endpoint $\sigma(1) = \sqrt{(\sigma^2-1)/(2\ln\sigma)}$ setting the maximum noise scale realised during training. Note that

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the hyperparameter σ is *not* the terminal noise scale $\sigma_{\max}$. The two are related by $\sigma_{\max} = \sigma(1)$.

B. Reverse-time generation

Generative sampling starts from the tractable terminal distribution(Gaussian distribution), drawing $\phi_{t_{\max}} \sim p_{t_{\max}}$, and then evolves the field backward in time to obtain a sample from $p_0 = p_{\text{data}}$. This backward evolution is not the inversion of an individual noisy trajectory, but a reverse-time stochastic process whose marginal densities undo the forward noising. Anderson [5] showed that the time-reversal of an Itô SDE is itself an Itô SDE. Applied to Eq. (1), this gives [2]

$$\mathrm{d}\phi \;=\; \big[f(\phi,t) - g^2(t)\,\nabla_\phi \log p_t(\phi)\big]\,\mathrm{d}t \;+\; g(t)\,\mathrm{d}\bar{w}, \quad (7)$$

where Eq. (7) is understood as being integrated in the reverse direction from $t = t_{\max}$ down to $t = 0$, so that $\mathrm{d}t < 0$ is an infinitesimal negative time step, and $\mathrm{d}\bar{w}$ denotes an independent standard Wiener increment at each such step. Song *et al.* [2] further established the existence of a deterministic probability-flow ODE

$$\frac{\mathrm{d}\phi}{\mathrm{d}t} \;=\; f(\phi,t) - \tfrac{1}{2}\,g^2(t)\,\nabla_\phi \log p_t(\phi) \quad (8)$$

whose marginal density at every $t \in [0, t_{\max}]$ coincides with that of the forward SDE Eq. (1) and the reverse SDE Eq. (7). Concretely, an ensemble initialised at $\phi_0 \sim p_{\text{data}}$ and evolved forward in t under Eq. (1), or initialised at $\phi_{t_{\max}} \sim p_{t_{\max}}$ and evolved backward in t under either Eq. (7) or Eq. (8), produces the same marginal $p_t(\phi)$ at every intermediate time slice, even though the three processes generate distinct individual trajectories. Being deterministic, Eq. (8) admits integration in either direction of t. For generative sampling we integrate it in the reverse direction from $t = t_{\max}$ down to $t = 0$, starting from $\phi_{t_{\max}} \sim p_{t_{\max}}$.

The crucial observation is that both reverse dynamics depend on the data distribution only through the time-dependent *score function*

$$s(\phi,t) \;=\; \nabla_\phi \log p_t(\phi), \quad (9)$$

a vector of the same dimension as ϕ. Replacing $s(\phi,t)$ in Eq. (7) and Eq. (8) with a neural approximation $s_\theta(\phi,t)$ yields the practical samplers

$$\mathrm{d}\phi \;=\; \big[f(\phi,t) - g^2(t)\,s_\theta(\phi,t)\big]\,\mathrm{d}t \;+\; g(t)\,\mathrm{d}\bar{w}, \quad (10)$$

$$\frac{\mathrm{d}\phi}{\mathrm{d}t} \;=\; f(\phi,t) - \tfrac{1}{2}\,g^2(t)\,s_\theta(\phi,t). \quad (11)$$

The training of s_θ is the subject of Sec. I C.

Although Eq. (7) and Eq. (8) have identical exact marginals, their discretisations with an imperfect s_θ behave very differently at the level of individual trajectories. The reverse SDE Eq. (10) is stochastic and non-invertible, but its injected noise acts as a Langevin-like corrector that damps accumulated score errors at each step [6]. The probability-flow ODE Eq. (11) is deterministic and invertible, enabling likelihood evaluation via the continuous change-of-variables formula and latent-space interpolation. It also admits higher-order numerical solvers such as DPM-Solver [7] or the Heun integrator of EDM [6], which reach high-quality generation in 10–35 function evaluations.

The choice of sampler therefore involves a trade-off. A deterministic ODE integrator faithfully reproduces any systematic bias present in s_θ, whereas the stochasticity of the reverse SDE provides a relaxation mechanism towards the true marginal at every time slice. For applications demanding high *spectral fidelity*, in particular when the two-point function $G(p)$ of generated configurations is used as a physically meaningful quality metric (as in the propagator study of Sec. IV), the reverse SDE, or a predictor-corrector hybrid that periodically injects Langevin noise into an ODE trajectory [2], therefore tends to produce more faithful behaviour.

C. Score matching

The reverse samplers Eq. (10) and Eq. (11) require the score $s(\phi,t) = \nabla_\phi \log p_t(\phi)$ at every noise level t. The marginal is itself an integral over the unknown data distribution,

$$p_t(\phi) \;=\; \int p_{\text{data}}(\phi_0)\,p_{0t}(\phi \mid \phi_0)\,\mathrm{d}\phi_0, \quad (12)$$

so $\nabla_\phi \log p_t$ is not available as a direct training target. *Denoising score matching* [8] sidesteps this by fitting $s_\theta(\phi,t)$ to the analytically known score of the *conditional* distribution $p_{0t}(\phi_t \mid \phi_0)$, minimising

$$\mathcal{L}(\theta) \;=\; \mathbb{E}_{t,\,\phi_0,\,\phi_t|\phi_0}\Big[\lambda(t)\,\big\|s_\theta(\phi_t,t) - \nabla_{\phi_t}\log p_{0t}\big(\phi_t \,\big|\, \phi_0\big)\big\|^2\Big], \quad (13)$$

with $\lambda(t) > 0$ a weighting function, $t \sim \text{Uniform}(0, t_{\max})$, $\phi_0 \sim p_{\text{data}}$, and $\phi_t \sim p_{0t}(\cdot \mid \phi_0)$. A key identity [8] shows that Eq. (13) and the (intractable) explicit score-matching loss $\mathbb{E}_{t,\phi_t}\big[\lambda(t)\|s_\theta(\phi_t,t) - \nabla_{\phi_t}\log p_t(\phi_t)\|^2\big]$ differ only by an additive θ-independent constant. The global minimiser of Eq. (13) is therefore the *marginal* score $\nabla_{\phi_t}\log p_t$, even though the target in Eq. (13) is only the conditional score. Because $p_{0t}(\phi_t \mid \phi_0)$ is Gaussian with known mean and variance (Eq. (3) and Eq. (5)), this target is elementary and the objective is directly tractable. Equivalent reparameterisations predict the injected noise ϵ (DDPM's "ϵ-prediction" [1]) or the clean signal ϕ_0.

For the VE schedule adopted in this work, the conditional score reads

$$\nabla_{\phi_t}\log p_{0t}\big(\phi_t \,\big|\, \phi_0\big) \;=\; -\frac{\phi_t - \phi_0}{\sigma^2(t)} \;=\; -\frac{\epsilon}{\sigma(t)}, \quad (14)$$

having used the reparameterisation $\phi_t = \phi_0 + \sigma(t)\,\epsilon$ with $\epsilon \sim \mathcal{N}(0, \mathbb{1})$. The standard weighting $\lambda(t) = \sigma^2(t)$ equalises the signal-to-noise ratio along the diffusion trajectory. Substituting Eq. (14) into Eq. (13) with this weighting reduces the objective to

$$\mathcal{L}(\theta) \;=\; \mathbb{E}_{t,\,\phi_0,\,\epsilon}\Big[\big\|\sigma(t)\, s_\theta(\phi_t, t) + \epsilon\big\|^2\Big], \qquad (15)$$

the loss minimised in all subsequent training runs.

Since $p_0 = p_{\rm data}$ by construction, the marginal satisfies $p_t \to p_{\rm data}$ as $t \to 0$, and a converged network satisfies

$$s_\theta(\phi, t) \;\xrightarrow[t\to 0]{}\; \nabla_\phi \log p_{\rm data}(\phi) \;=\; -\nabla_\phi S[\phi], \qquad (16)$$

where the final equality uses the Boltzmann form $p_{\rm data}(\phi) \propto e^{-S[\phi]}$ of the target distribution. The small-t limit of the learned score therefore reproduces the classical force $-\nabla_\phi S[\phi]$ of the action, as a byproduct of denoising-based training. The precision of this limit, and the sensitivity of the reverse samplers Eq. (10) and Eq. (11) to its residual error, underlie the diagnostics developed in Sec. IV A and Sec. IV B.

II. CRITICAL POINT OF 2D ϕ^4 THEORY FROM WOLFF + FOURIER-ACCELERATED HMC

We consider the scalar ϕ^4 theory on a two-dimensional $L \times L$ square lattice with periodic boundary conditions. The lattice action reads

$$S[\phi] = \sum_x \Big[-2\kappa \sum_{\mu=1}^{2} \phi_x\, \phi_{x+\hat\mu} + \phi_x^2 + \lambda\big(\phi_x^2 - 1\big)^2\Big], \quad (17)$$

where κ is the hopping parameter and λ the quartic self-coupling. In two dimensions we set $\lambda = 0.022$. The model undergoes a $\mathbb{Z}_2$-symmetry-breaking phase transition at a critical value $\kappa_c(\lambda)$, which belongs to the 2D Ising universality class.

A. Sampling algorithm

Configurations are generated by a hybrid algorithm that alternates Fourier-accelerated Hybrid Monte Carlo (FA-HMC) updates with Wolff cluster flips.

a. FA-HMC. The standard HMC suffers from critical slowing down because the mass matrix in momentum space does not match the spectrum of the lattice operator. In Fourier-accelerated HMC, one introduces a momentum-dependent mass matrix in Fourier space,

$$M(\hat p) = \hat p^{\,2} + m_{\rm eff}^2, \quad \hat p^{\,2} = \sum_{\mu=1}^{D} 4\sin^2\!\Big(\frac{\pi k_\mu}{L}\Big), \qquad (18)$$

where $m_{\rm eff}^2$ is estimated from the variance of Fourier modes during a warm-up phase and subsequently adapted in a series of windows. The conjugate momenta are sampled as $\pi \sim \mathcal{N}(0, M)$ and the kinetic energy is $K = \frac{1}{2}\pi^\top M^{-1}\pi$, both evaluated efficiently via FFT. The leapfrog integrator uses the velocity $v = M^{-1}\pi$ for the position update, ensuring that long-wavelength modes propagate with a step size comparable to that of short-wavelength modes. The step size ε is tuned via dual averaging to maintain an acceptance rate of $\sim 75\%$.

b. Wolff cluster flip. After each FA-HMC trajectory, we perform $n_{\rm Wolff}$ Wolff cluster updates that target the global $\mathbb{Z}_2$ mode. A cluster is grown from a random seed site by adding neighboring site y to the cluster with bond probability

$$P_{\rm bond}(x, y) = 1 - \exp\!\big(-4\kappa\, \phi_x \phi_y\big), \qquad (19)$$

whenever $\phi_x\phi_y > 0$. All sites in the cluster are then reflected: $\phi_x \to -\phi_x$. This step efficiently decorrelates the magnetization and drastically reduces critical slowing down for the order parameter.

B. Observables

From the spatially averaged magnetization per configuration, $\bar\phi = \frac{1}{V}\sum_x \phi_x$ with $V = L^2$, we measure:

1. **Order parameter:** $\langle|\bar\phi|\rangle$.
2. **Magnetic susceptibility:** $\chi = V\big(\langle|\bar\phi|^2\rangle - \langle|\bar\phi|\rangle^2\big)$.
3. **Binder cumulant:** $U_L = 1 - \dfrac{\langle|\bar\phi|^4\rangle}{3\,\langle|\bar\phi|^2\rangle^2}$.

Statistical errors are estimated by the jackknife method.

C. Results

Fig. 1 shows the order parameter $\langle|\bar\phi|\rangle$ as a function of κ for $L = 16, 32, 64, 128, 256$. A clear crossover from the broken phase ($\langle|\bar\phi|\rangle \to 0$) to the symmetric phase ($\langle|\bar\phi|\rangle > 0$) is observed, with the transition sharpening as L increases.

The magnetic susceptibility χ is shown in Figs. 2 and 3. The peak of χ grows with L and shifts toward the critical κ_c, the expected finite-size scaling $\chi_{\max} \sim L^{\gamma/\nu}$ of the 2D Ising universality class ($\gamma/\nu = 7/4$).

The Binder cumulant U_L is displayed in Fig. 4. In the thermodynamic limit, $U_L \to 2/3$ in the broken phase and $U_L \to 0$ in the symmetric phase. Curves for different L cross at the critical point κ_c, which is a powerful method for locating the phase transition independent of critical exponents.

Fig. 5 shows the Binder crossing analysis. From the intersection of the U_L curves for different lattice-size pairs, we extract the critical hopping parameter

$$\kappa_c(\lambda = 0.022) \approx 0.27088, \qquad (20)$$

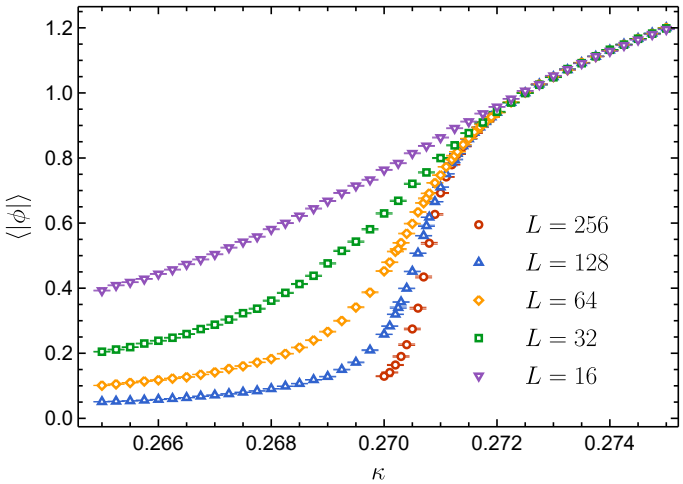

FIG. 1. Order parameter $\langle|\bar{\phi}|\rangle$ as a function of the hopping parameter κ for lattice sizes $L = 16, 32, 64, 128, 256$ at $\lambda = 0.022$. The transition becomes sharper with increasing volume.

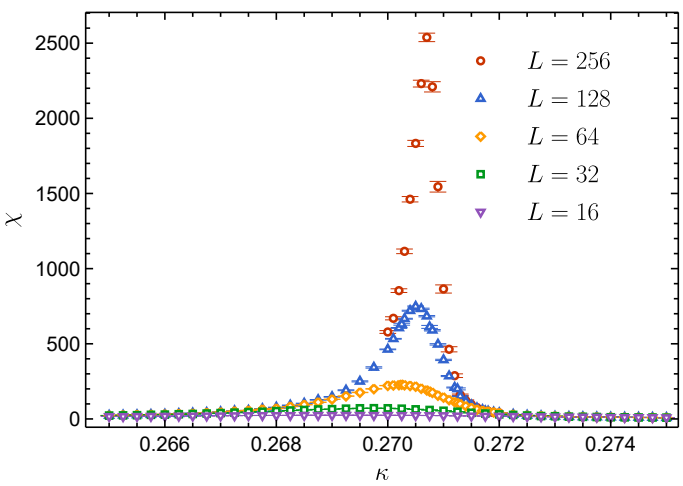

FIG. 2. Magnetic susceptibility χ as a function of κ for different lattice sizes.

III. CRITICAL POINT OF 3D ϕ^4 THEORY FROM WOLFF + FA-HMC

We extend the analysis to the three-dimensional ϕ^4 theory on an L^3 cubic lattice. The lattice action takes the same form as Eq. (17) with the sum over nearest neighbors running over $D = 3$ directions. In three dimensions, we set the quartic coupling to $\lambda = 0.9$. The $\mathbb{Z}_2$-breaking transition in 3D belongs to the 3D Ising universality class, with critical exponents $\nu \approx 0.63$.

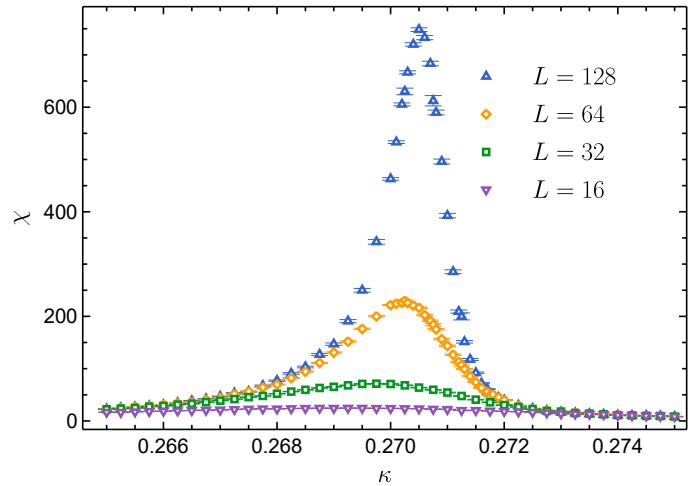

FIG. 3. Magnetic susceptibility χ (alternative representation) as a function of κ.

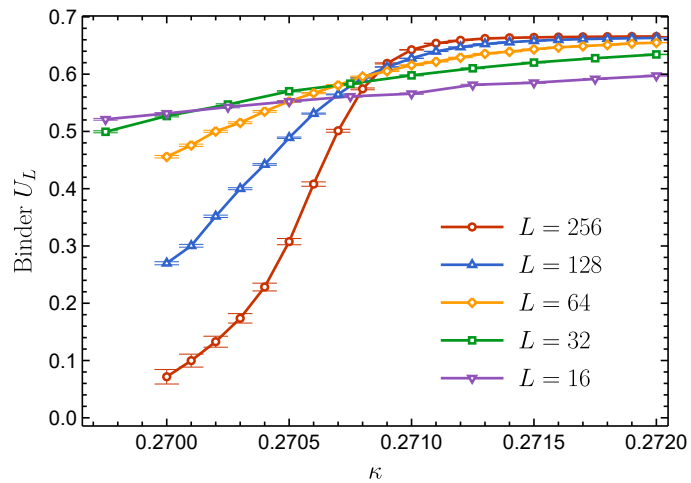

FIG. 4. Binder cumulant U_L as a function of κ for $L = 16, 32, 64, 128, 256$. The crossing of curves at different L determines the critical point κ_c.

A. Simulation parameters

B. Results

Fig. 6 shows the order parameter $\langle|\bar{\phi}|\rangle$ as a function of κ for $L = 8, 16, 32, 64$. The crossover from the symmetric to the broken phase sharpens with increasing volume, analogous to the 2D case.

The magnetic susceptibility χ is displayed in Fig. 7. The peak height grows with L and its position converges toward κ_c.

The Binder cumulant U_L is shown in Fig. 8. Curves for different L cross in a narrow region, pinpointing the critical hopping parameter. From the intersection of the $L = 32$ and $L = 64$ curves, we estimate

$$\kappa_c(\lambda = 0.9) \approx 0.19225, \tag{21}$$

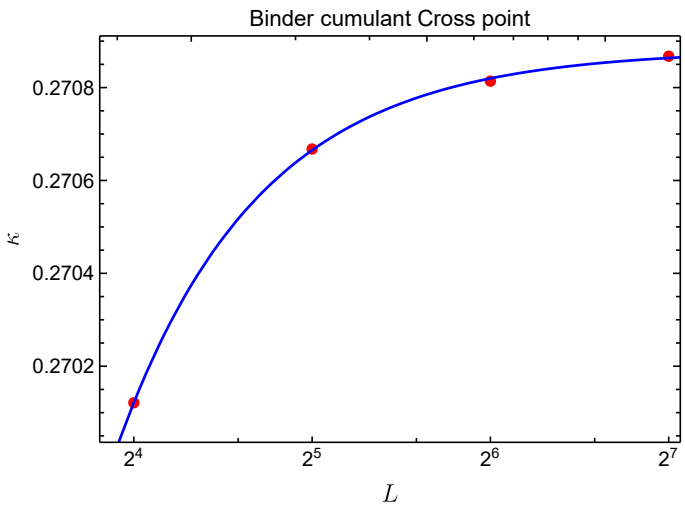

FIG. 5. Binder crossing analysis for the determination of κ_c. The intersection of different lattice-size pairs yields $\kappa_c \approx 0.27088$.

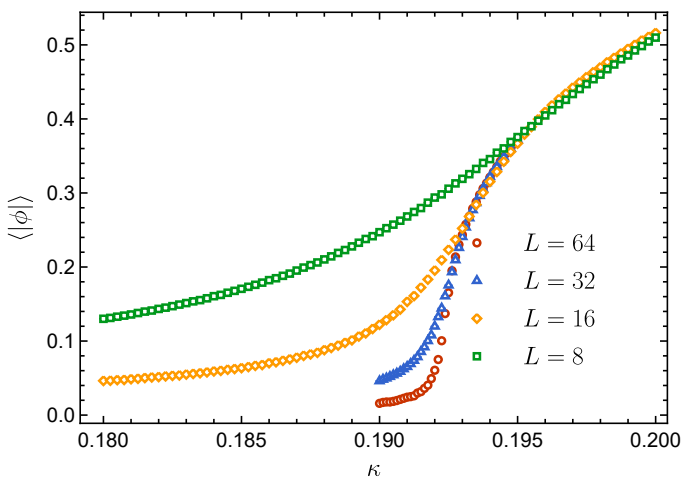

FIG. 6. Order parameter $\langle|\bar{\phi}|\rangle$ vs. κ for the 3D ϕ^4 theory ($\lambda = 0.9$) at lattice sizes $L = 8$, 16, 32, 64.

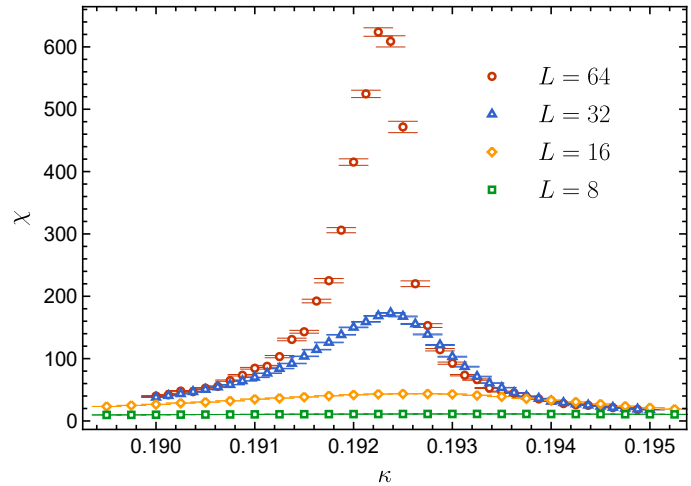

FIG. 7. Magnetic susceptibility χ vs. κ for the 3D theory at different lattice sizes.

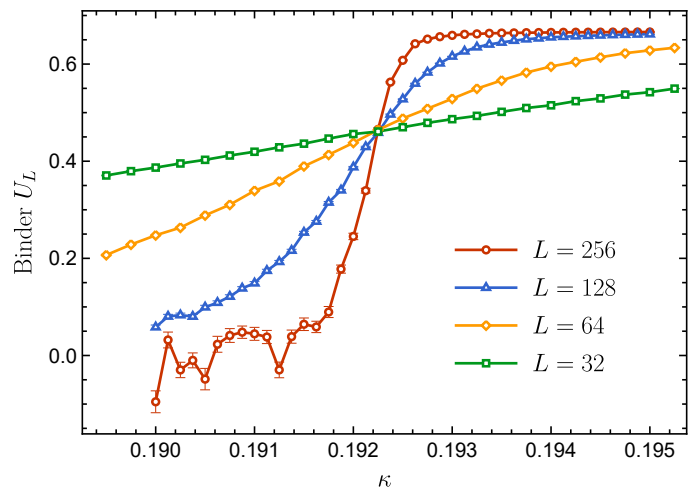

FIG. 8. Binder cumulant U_L vs. κ for the 3D ϕ^4 theory at $L = 8$, 16, 32, 64. The crossing of curves for different L determines $\kappa_c \approx 0.19225$.

consistent with $U_L^* \approx 2/3$ at the crossing, as expected for the 3D Ising universality class.

IV. MOMENTUM-SPACE PROPAGATOR FROM DIFFUSION MODELS

Having established the critical points for the 2D and 3D ϕ^4 theories via Wolff + FA-HMC, we now test whether a score-based diffusion model can faithfully reproduce the momentum-space propagator $G(p)$. We train a noise-conditional score network (NCSN++) on HMC-generated configurations at $L = 128$ and $\lambda = 0.022$, then generate independent samples by reverse-diffusion and compare the radially-averaged propagator

$$G(|p|) = \left\langle |\tilde{\phi}(p)|^2 \right\rangle_{\text{radial}} \tag{22}$$

against the HMC reference, where $\tilde{\phi}(p)$ is the discrete Fourier transform of a configuration. Errors are estimated via bootstrap.

We consider three representative values of the hopping parameter that span the phase diagram:

- $\kappa = 0.26$, in the symmetric phase;
- $\kappa = 0.2705 \approx \kappa_c$, near the critical point;
- $\kappa = 0.28$, in the broken phase.

A. Acceptance rate as a training diagnostic

A natural quantitative measure of how well a diffusion model has learned the target distribution is

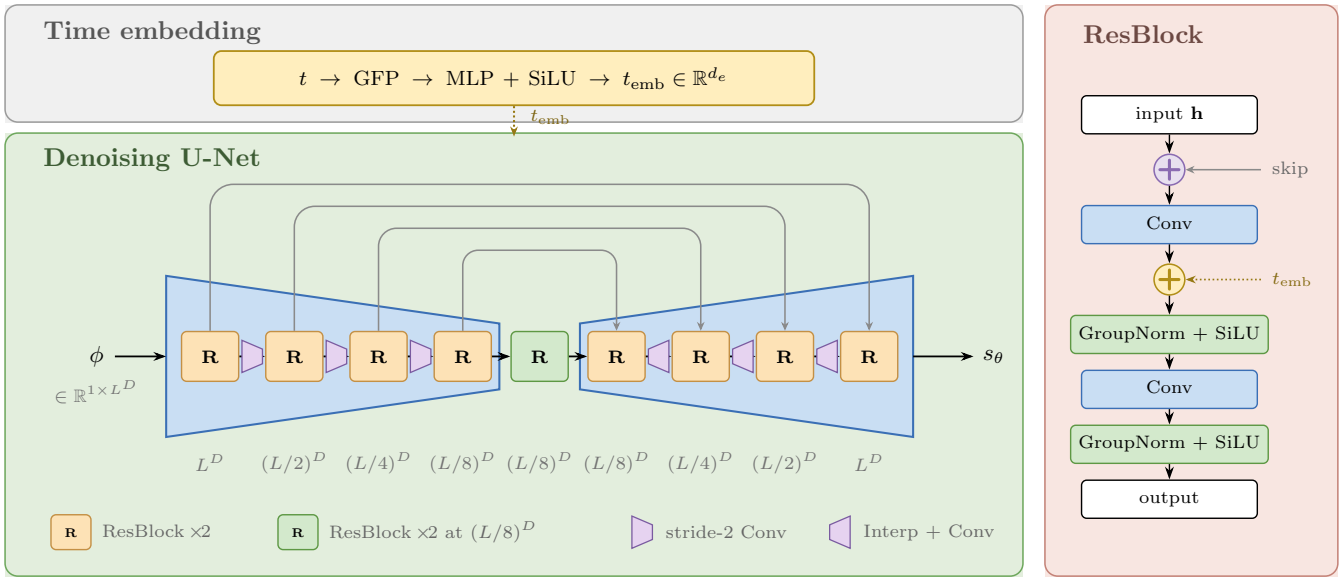

FIG. 9. Architecture of the NCSN++ score network used for both 2D and 3D ϕ^4 lattice field theory. **(a)** U-Net overview. The encoder (blue) consists of four ResBlock pairs at progressively coarser resolutions, separated by stride-2 convolutions (purple) that halve the spatial dimensions. The decoder (orange) mirrors this structure, upsampling via nearest-neighbor interpolation followed by a convolution (purple). Each encoder level provides two skip connections (from both sub-blocks) to the corresponding decoder level via channel concatenation ("concat ×2"). The diffusion time t is encoded by Gaussian Fourier projection (GFP) and a learned MLP, then injected additively into every ResBlock. The final output is divided by $\sigma(t)$. **(b)** Internal structure of a single ResBlock; no residual connection is used. All convolutions use circular padding to enforce periodic boundary conditions. For 2D fields all convolutions are `Conv2d`; for 3D they become `Conv3d` ($D = 3$).

the Metropolis–Hastings (MH) acceptance rate of a Metropolis-Adjusted Langevin Algorithm (MALA) [9] that uses the score network as its proposal drift.

In the reverse-diffusion framework, the score network $s_\theta(\phi, t) \approx \nabla_\phi \log p_t(\phi)$ is trained to approximate the score of the noised distribution p_t. At small diffusion time $t \to 0$, $p_t \to p_0 \propto e^{-S}$ and therefore $s_\theta(\phi, t) \to -\nabla_\phi S(\phi)$. From the current state ϕ_τ, the MALA proposal is

$$\psi_{\tau+1} = \phi_\tau + h\, s_\theta(\phi_\tau,\, t_{\text{mh}}) + \sqrt{2h}\,\eta, \tag{23}$$

where h is the step size, t_{mh} the diffusion time at which the score is evaluated, and $\eta \sim \mathcal{N}(0, \mathbb{1})$. Conditioned on ϕ_τ, the proposal is a Gaussian with mean $\phi_\tau + h\, s_\theta(\phi_\tau, t_{\text{mh}})$ and variance $2h\, I_n$, so the proposal kernel has the explicit form

$$q(\psi_{\tau+1}|\phi_\tau) = \frac{1}{(4\pi h)^{n/2}} \exp\Big(-\frac{1}{4h}\big\|\psi_{\tau+1} - \phi_\tau - h\, s_\theta(\phi_\tau, t_{\text{mh}})\big\|^2\Big), \tag{24}$$

where $n = L^D$ is the number of degrees of freedom and $p(\phi) \propto e^{-S(\phi)}$ is the known physical distribution. The accept-reject step is

$$\phi_{\tau+1} = \begin{cases} \psi_{\tau+1} & \text{with prob. } \min\Big\{1,\, \frac{p(\psi_{\tau+1})\, q(\phi_\tau|\psi_{\tau+1})}{p(\phi_\tau)\, q(\psi_{\tau+1}|\phi_\tau)}\Big\}, \\ \phi_\tau & \text{otherwise.} \end{cases} \tag{25}$$

The transition probability ratio can be computed explicitly as

$$\frac{p(\psi_{\tau+1})\, q(\phi_\tau|\psi_{\tau+1})}{p(\phi_\tau)\, q(\psi_{\tau+1}|\phi_\tau)} = \exp\Big(-S(\psi_{\tau+1}) - \tfrac{1}{4h}\big\|\phi_\tau - \psi_{\tau+1} - h\, s_\theta(\psi_{\tau+1}, t_{\text{mh}})\big\|^2 + S(\phi_\tau) + \tfrac{1}{4h}\big\|\psi_{\tau+1} - \phi_\tau - h\, s_\theta(\phi_\tau, t_{\text{mh}})\big\|^2\Big). \tag{26}$$

Substituting the proposal (23), i.e. $\psi_{\tau+1}-\phi_\tau-h\,s_\theta(\phi_\tau)=\sqrt{2h}\,\eta$, the log acceptance probability simplifies to

$$\log\alpha = \big[S(\phi_\tau)-S(\psi_{\tau+1})\big]+\tfrac{1}{2}\|\eta\|^2-\tfrac{1}{4h}\big\|\sqrt{2h}\,\eta+h\big[s_\theta(\phi_\tau,t_{\rm mh})+s_\theta(\psi_{\tau+1},t_{\rm mh})\big]\big\|^2, \tag{27}$$

which is the form used in practice. The MH step guarantees that, regardless of the quality of s_θ, the chain targets the correct distribution e^{-S}. The acceptance rate $\langle\min(1,e^{\log\alpha})\rangle$ therefore measures how closely the score-network proposal matches the true Langevin drift $-\nabla_\phi S$: a well-trained model yields a high acceptance rate, while a poorly trained one is rejected more often.

a. Parameter calibration. Two parameters must be chosen: the step size h and the diffusion time $t_{\rm mh}$. The calibration criterion is the *diagnostic power*, defined as the difference in acceptance rate between a late-epoch (well-trained) and an early-epoch (poorly trained) checkpoint.

b. Step-size scaling. For MALA on a d-dimensional i.i.d. product target $\pi(x)=\prod_i f(x_i)$, Roberts and Rosenthal [9] proved that the optimal proposal variance scales as $h\sim d^{-1/3}$, yielding an optimal acceptance rate of 0.574. For a D-dimensional lattice with $d=L^D$ sites, this would give $h\sim L^{-D/3}$, i.e. $h\sim L^{-2/3}$ in 2D and $h\sim L^{-1}$ in 3D. However, the ϕ^4 action contains nearest-neighbour couplings ($-2\kappa\sum_{\langle x,y\rangle}\phi_x\phi_y$) that strongly violate the i.i.d. assumption. The total action change for a global proposal scales as $\Delta S\sim d\cdot h$, so maintaining $\Delta S\sim O(1)$ for reasonable acceptance requires the RWM-type scaling $h\sim 1/d=1/L^D$ rather than the MALA-optimal $d^{-1/3}$. Empirically, $h\sim L^{-D/3}$ produces a step size that is far too large in both 2D and 3D. We therefore adopt $h=c/L^D$ throughout, i.e. $h=c/L^2$ in 2D and $h=c/L^3$ in 3D, with the coefficient c calibrated independently for each dimensionality.

We find that the diagnostic power is maximised at $t_{\rm mh}=10^{-4}$, which is much smaller than values typically used in sampling ($t\sim10^{-2}$). As shown in Fig. 11, at large $t_{\rm mh}\gtrsim 5\times10^{-3}$ the gap reverses sign: the score at large t approximates the score of a heavily smeared distribution $p_t\neq p_0$, so a well-trained model's proposal is rejected more often by the exact action than a poorly trained one whose drift is weak and effectively random. This reversal makes large $t_{\rm mh}$ unsuitable as a diagnostic. For the step size in 2D, Fig. 10 shows that $c=0.2$ (i.e. $h=0.2/L^2$) maximises the diagnostic power while keeping the late-epoch acceptance rate in the informative range 50–80%.

With the calibrated parameters $c=0.2$ and $t_{\rm mh}=10^{-4}$, we measure the MALA acceptance rate using $N=1024$ reference configurations drawn from the training data and a single MH step per configuration. Using a single MH step avoids the warm-up bias that arises with multiple steps (where the MH chain migrates toward lower-action regions, artificially inflating the acceptance rate) and maximises statistical independence across samples.

TABLE I. Average MALA acceptance rate over epochs 49–249 (early training), with $c=0.2$, $t_{\rm mh}=10^{-4}$, $N=1024$ and a single MH step.

L	$\kappa=0.26$	$\kappa=0.2705$	$\kappa=0.28$
8	0.152	0.136	0.129
16	0.082	0.228	0.031
32	0.207	0.205	0.297
64	0.288	0.375	0.300
128	0.336	0.325	0.296

c. Acceptance rate during training. Fig. 12 shows the acceptance rate as a function of training epoch for all three κ values at $L=128$. In all three cases, the acceptance rate starts near zero for the untrained model (epoch 49) and rises rapidly during the first ∼500 epochs, eventually fluctuating in the range 60–90%. The substantial epoch-to-epoch fluctuations reflect the stochastic nature of both the score-network training and the finite-sample MALA estimator, but the overall upward trend provides a clear signal that the score network progressively learns the correct Langevin drift.

d. Acceptance rate across lattice sizes. Tabs. I and II summarise the average MALA acceptance rate for all trained models at two stages of training: an early window (epochs 49–249) and a late window (epochs 3999–4999). In all cases, the late-epoch acceptance rate is substantially higher than the early-epoch value, confirming that the diagnostic tracks training progress. The late-epoch acceptance rate increases mildly with lattice size, from ∼ 63–71% at $L=8$ to ∼ 75–82% at $L=64$–128. This trend is consistent with the $h=c/L^2$ scaling being slightly more conservative than the true optimal scaling for the correlated ϕ^4 target: the optimal exponent likely lies between $L^{-2/3}$ (i.i.d. limit [9]) and L^{-2} (our choice), so at larger L the step size is increasingly conservative and the acceptance rate rises accordingly. Despite this mild L-dependence, the acceptance rate remains a useful diagnostic at all lattice sizes tested.

B. Direct score quality diagnostic

While the MALA acceptance rate provides an effective indirect measure of score quality, it depends on the step size h and evaluation time $t_{\rm mh}$, requiring careful calibration (Fig. 10, Fig. 11). A more direct diagnostic is to compare the model output $s_\theta(\phi,t)$ against the true gra-

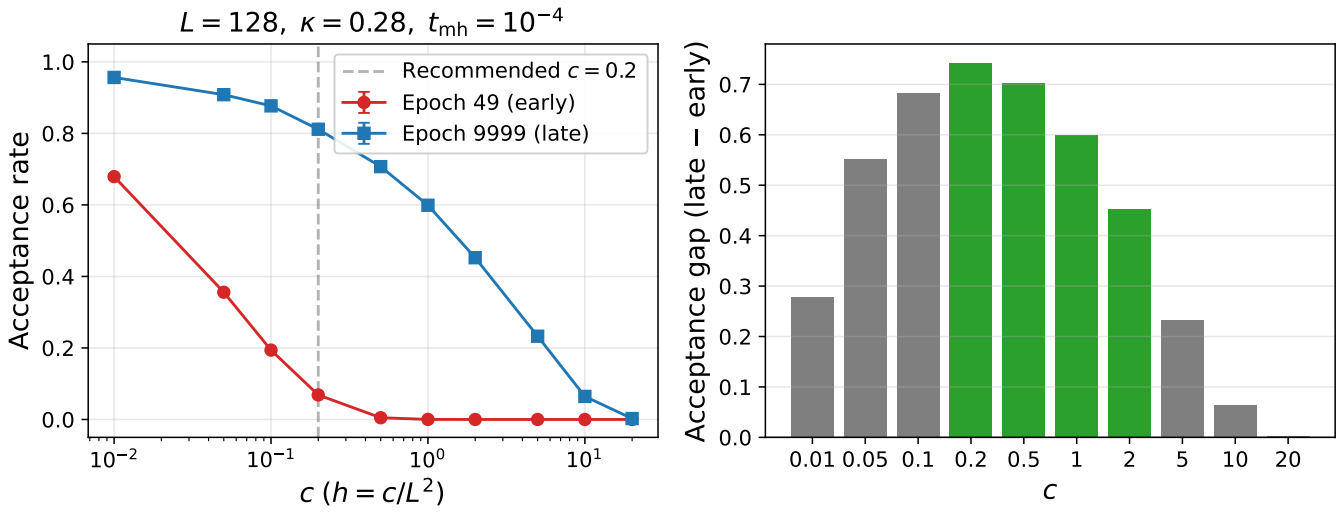

FIG. 10. Calibration of the MALA step size $h = c/L^2$ at $L = 128$, $\kappa = 0.28$, with $t_{\rm mh} = 10^{-4}$, $N = 1024$ reference configurations and a single MH step. Left: acceptance rate vs. c for the early-epoch (epoch 49, poorly trained) and late-epoch (epoch 9999, well-trained) checkpoints. Right: diagnostic power (acceptance gap between late and early checkpoints). Green bars indicate the "useful" range where the late-epoch acceptance rate is between 30% and 85%.

FIG. 11. Calibration of the evaluation time $t_{\rm mh}$ at $L = 128$, $\kappa = 0.28$, with $c = 0.2$, $N = 1024$ and a single MH step. Left: acceptance rate vs. $t_{\rm mh}$ for early (epoch 49) and late (epoch 9999) checkpoints. Right: diagnostic power. The gap reverses sign for $t_{\rm mh} \gtrsim 5 \times 10^{-3}$, where the score network evaluates the gradient of a heavily smeared distribution rather than the physical action.

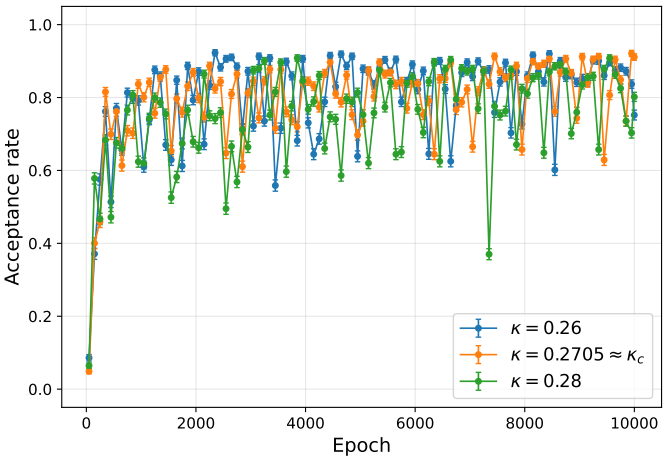

FIG. 12. MALA acceptance rate vs. training epoch for the 2D ϕ^4 model at $L = 128$, with $c = 0.2$, $t_{\rm mh} = 10^{-4}$, $N = 1024$ and a single MH step, for three values of κ spanning the phase diagram.

dient $-\nabla_\phi S(\phi)$ at a small diffusion time t, without any Metropolis–Hastings sampling.

For each reference configuration ϕ drawn from the training data, we evaluate the model score $s_\theta(\phi, t_{\rm eval})$

TABLE II. Same as Tab. I but averaged over epochs 3999–4999 (late training).

L	$\kappa = 0.26$	$\kappa = 0.2705$	$\kappa = 0.28$
8	0.635	0.715	0.627
16	0.683	0.641	0.693
32	0.680	0.753	0.683
64	0.827	0.818	0.747
128	0.790	0.792	0.749

at $t_{\rm eval} = 10^{-4}$ and compute the cosine similarity

$$C = \frac{\langle s_\theta, -\nabla S\rangle}{\|s_\theta\|\ \|\nabla S\|}, \tag{28}$$

the relative mean-squared error $\|s_\theta - (-\nabla S)\|^2/\|\nabla S\|^2$, and the magnitude ratio $\|s_\theta\|/\|\nabla S\|$, averaged over $N = 512$ configurations. The choice $t_{\rm eval} = 10^{-4}$ (noise level $\sigma \approx 0.01$, or 0.5% of the data range in normalised space) ensures that $p_t \approx p_0$ to high accuracy while the score estimate remains numerically stable. A larger evaluation time $t = 0.01$ ($\sigma \approx 0.10$) introduces non-negligible Gaussian smoothing that systematically reduces the magnitude ratio below unity even for a perfectly trained model, making it a less informative diagnostic; a smaller time $t = 10^{-5}$ ($\sigma \approx 0.003$) enters the regime where division by $\sigma(t)$ in the network output amplifies numerical noise.

Fig. 13 shows all three metrics as a function of training

TABLE III. Average cosine similarity C and magnitude ratio $\|s_\theta\|/\|\nabla S\|$ over epochs 49–249 (early training), evaluated at $t = 10^{-4}$ with $N = 512$ reference configurations.

	Cosine similarity C			Magnitude ratio		
L	κ=0.26	0.2705	0.28	κ=0.26	0.2705	0.28
8	0.987	0.971	0.974	2.102	2.018	1.942
16	0.985	0.977	0.976	2.046	1.716	2.042
32	0.981	0.981	0.986	1.704	1.647	1.568
64	0.992	0.985	0.990	1.581	1.401	1.408
128	0.994	0.994	0.989	1.512	1.472	1.459

TABLE IV. Same as Tab. III but averaged over epochs 3999–4999 (late training).

	Cosine similarity C			Magnitude ratio		
L	κ=0.26	0.2705	0.28	κ=0.26	0.2705	0.28
8	0.988	0.991	0.990	1.282	1.148	1.267
16	0.989	0.989	0.993	1.119	1.158	1.052
32	0.988	0.993	0.990	1.053	1.014	0.953
64	0.996	0.996	0.996	0.968	0.972	0.949
128	0.997	0.997	0.998	0.907	0.938	0.950

epoch for $L = 128$ at all three κ values. The cosine similarity rises from ~ 0.99 at epoch 49 to ~ 0.998 at late training; the relative MSE drops by more than an order of magnitude; and the magnitude ratio converges toward unity from above, starting at ~ 1.5 and reaching ~ 0.9–1.0 by epoch 5000. These trends are consistent with the monotonic improvement observed in the MALA acceptance rate (Fig. 12) and confirm that the score network progressively learns both the direction and the magnitude of $-\nabla S$.

Tabs. III and IV summarise the average cosine similarity and magnitude ratio across all lattice sizes and κ values, in the same early and late training windows used for the acceptance rate (Tabs. I and II). In the early window (epochs 49–249), the magnitude ratio is ~ 1.4–2.1, indicating that the untrained model significantly overshoots the true gradient magnitude. By late training (epochs 3999–4999), the magnitude ratio converges to 0.9–1.1 for $L \geq 32$, confirming quantitative accuracy of the learned score. For $L = 8$ and 16 a slight residual overshoot ($\sim$ 1.1–1.3) persists, likely reflecting the smaller number of degrees of freedom where the model has less room to average out local fluctuations.

C. Training loss for 3D ϕ^4

We train the NCSN++ model on Wolff + FA-HMC configurations of the 3D theory at $L = 64$ and $\lambda = 0.9$ for three hopping parameters: $\kappa = 0.18$ (symmetric phase), $\kappa = 0.1923 \approx \kappa_c$ (near the critical point), and $\kappa = 0.2$ (broken phase). The score-matching loss is decomposed by the diffusion time t into a UV component ($t < 0.2$, probing short-distance modes), an intermediate component ($0.2 < t < 0.8$), and an IR component ($t > 0.8$, probing long-distance modes).

Fig. 14 shows the training loss evolution up to 20000 epochs. The total loss (panel a) converges for all three κ values, with a clear hierarchy: the symmetric-phase model ($\kappa = 0.18$) has the highest loss plateau, while the broken-phase model ($\kappa = 0.2$) converges to the lowest value. The decomposition reveals that the IR loss (panel d) is the dominant contribution and the most sensitive to κ: the symmetric phase shows the largest IR loss, reflecting the challenge of learning long-wavelength modes when the correlation length is large but finite. The mid and UV components (panels b, c) show a similar ordering, confirming that the phase-dependent difficulty extends across all noise levels.

D. Acceptance rate for 3D ϕ^4

We apply the same MALA acceptance-rate diagnostic to the 3D models at $L = 64$, $\lambda = 0.9$, now with $h = c/L^3$ and $N = 256$ reference configurations. Fig. 15 shows the step-size calibration: the optimal coefficient is again $c = 0.2$, the same value found in 2D. This consistency strongly supports the $h = c/L^D$ scaling: the coefficient c is universal across spatial dimensions once the volume factor L^D is correctly accounted for. The $t_{\rm mh}$ calibration (Fig. 16) with $c = 0.5$ confirms that $t_{\rm mh} = 10^{-4}$ maximises the diagnostic power, as in the 2D case.

Fig. 17 shows the acceptance rate vs. training epoch for all three 3D models with $c = 0.5$. The acceptance rates reach 60–90% at late epochs, comparable to the 2D results and confirming that the $h = c/L^D$ scaling yields a consistent diagnostic across dimensions.

a. Direct score quality for 3D. We apply the same direct score-quality diagnostic to the 3D models, evaluating the cosine similarity, relative MSE, and magnitude ratio at $t_{\rm eval} = 10^{-4}$ with $N = 64$ reference configurations. Fig. 18 shows the three metrics as a function of training epoch for all three κ values. The cosine similarity rises from ~ 0.96 at early training to ~ 0.995 at late training; the relative MSE drops by more than an order of magnitude; and the magnitude ratio converges to $\sim$ 0.97–1.00 from an initial overshoot of $\sim$ 1.4–1.5. These trends mirror the 2D results (Fig. 13) and confirm that the score network learns both the direction and the magnitude of $-\nabla S$ in three dimensions.

E. Quantitative propagator-mismatch metrics

To quantify how well the diffusion-model propagator $G_{\rm DM}(k)$ matches the reference $G_{\rm train}(k)$ across momentum bins, one could compare them via absolute deviation $\Delta G(k) = G_{\rm DM} - G_{\rm train}$ or relative deviation $\Delta G/G_{\rm train}$, but neither scales uniformly across the UV/IR range. The absolute deviation is dominated by a handful of

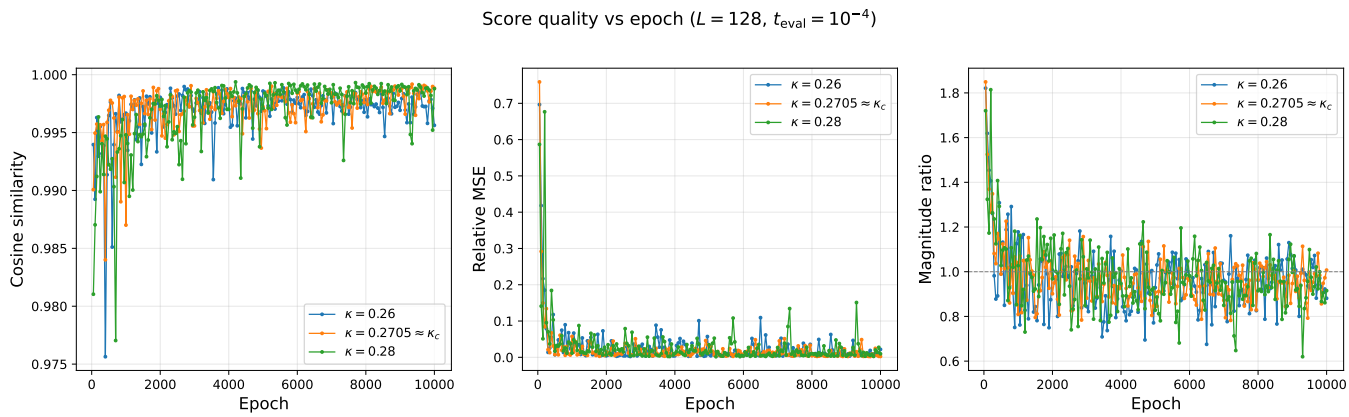

FIG. 13. Score quality vs. training epoch for the 2D ϕ^4 model at $L = 128$, evaluated at $t = 10^{-4}$ with $N = 512$ reference configurations. Left: cosine similarity. Centre: relative MSE. Right: magnitude ratio (dashed line indicates the ideal value of 1).

low-k bins where G is large, while the relative deviation grows artificially at high k where G is tiny and the mismatch contributes negligibly to the field variance. We instead use three complementary metrics, each with a sharp probabilistic or statistical interpretation.

a. (i) Per-mode Gaussian KL. Under translation invariance, any zero-mean Gaussian field measure diagonalises in momentum with covariance $G(k)$, and the modes become independent. The Kullback–Leibler divergence between the two Gaussian field distributions $p_{\text{train}} = \mathcal{N}(0, G_{\text{train}})$ and $p_{\text{DM}} = \mathcal{N}(0, G_{\text{DM}})$ therefore reduces to an additive sum over momentum modes:

$$D_{\text{KL}}(p_{\text{train}} \| p_{\text{DM}}) = \sum_k D_k, \qquad D_k = \tfrac{1}{2}\big[r_k - 1 - \log r_k\big], \quad r_k = \frac{G_{\text{train}}(k)}{G_{\text{DM}}(k)} \tag{29}$$

The function $f(r) = r - 1 - \log r$ vanishes at $r = 1$, is strictly convex, grows linearly as $r \to \infty$ (linear penalty for DM underestimating the variance) and only logarithmically as $r \to 0^+$ (weaker penalty for overestimation). For small relative deviations $r = 1 + \varepsilon$ one has $D_k \simeq \varepsilon^2/4$, so D_k reduces to the usual quadratic relative-deviation measure in the well-trained regime, but is far tamer than $(r-1)^2$ for large mismatches in the UV tail. Because D_k is dimensionless and additive, contributions from different $|k|$-bins (and hence UV vs IR halves) can be compared and summed on equal footing.

Crucially, (29) is the KL between two *Gaussian surrogates* determined by the two-point functions alone. When the underlying field theory is non-Gaussian (as in ϕ^4 with interactions), the metric captures the second-moment mismatch but is blind to higher-cumulant structure such as the connected four-point function $\langle \phi^4 \rangle_{\text{conn}}$; matching the two-point function is a necessary but not sufficient condition for distributional equivalence. The generalisation $D_{\text{KL}}(p \| q_{\text{Gauss}}) = D_{\text{KL}}(p_{\text{Gauss}} \| q_{\text{Gauss}}) + J(p)$, with $J(p) \geq 0$ the negentropy of the true distribution, makes the missing piece explicit.

b. (ii) Statistical z-score. Bootstrap-resampling the configurations yields uncertainty estimates $\sigma_{\text{DM}}(k)$ and $\sigma_{\text{train}}(k)$ on the two propagators. The standardised residual

$$z_k = \frac{G_{\text{DM}}(k) - G_{\text{train}}(k)}{\sqrt{\sigma^2_{\text{DM}}(k) + \sigma^2_{\text{train}}(k)}} \tag{30}$$

tests whether the per-bin discrepancy is statistically significant given the finite sample size. Bins with $|z_k| \lesssim 2$ are consistent with the reference at the 2σ level; persistent deviations with $|z_k| \gg 2$ across many bins signal a real bias rather than sampling noise. The root-mean-square $z_{\text{rms}} = \sqrt{\langle z_k^2 \rangle}$ provides a single scalar that saturates at $\mathcal{O}(1)$ when the mismatch is at the sample-noise floor and grows without bound when the bias dominates.

c. (iii) Phase-space weighted propagator difference. Expressed in the continuum limit, the field variance decomposes as $\langle \phi^2 \rangle = \int d^D k/(2\pi)^D \, G(k) \propto \int_0^\infty dk \, k^{D-1} G(k)$, so that each radial momentum shell contributes its propagator weighted by the phase-space measure k^{D-1}. Accordingly, the density

$$w_k = k^{D-1}\big[G_{\text{DM}}(k) - G_{\text{train}}(k)\big] \tag{31}$$

measures, up to an overall $(2\pi)^{-D}$ factor, how much each momentum shell contributes to the total mismatch in $\langle \phi^2 \rangle$. Integrating w_k over a UV or IR band directly quantifies the corresponding contribution to the two-point mismatch in position space, with physical units. Compared to D_k and z_k, w_k has a concrete physical interpretation but is not dimensionless, so the three metrics should be viewed as complementary rather than redundant.

d. Aggregate scalars. Each per-mode quantity above also admits a single-scalar summary. These scalars are convenient training diagnostics, one number per check-

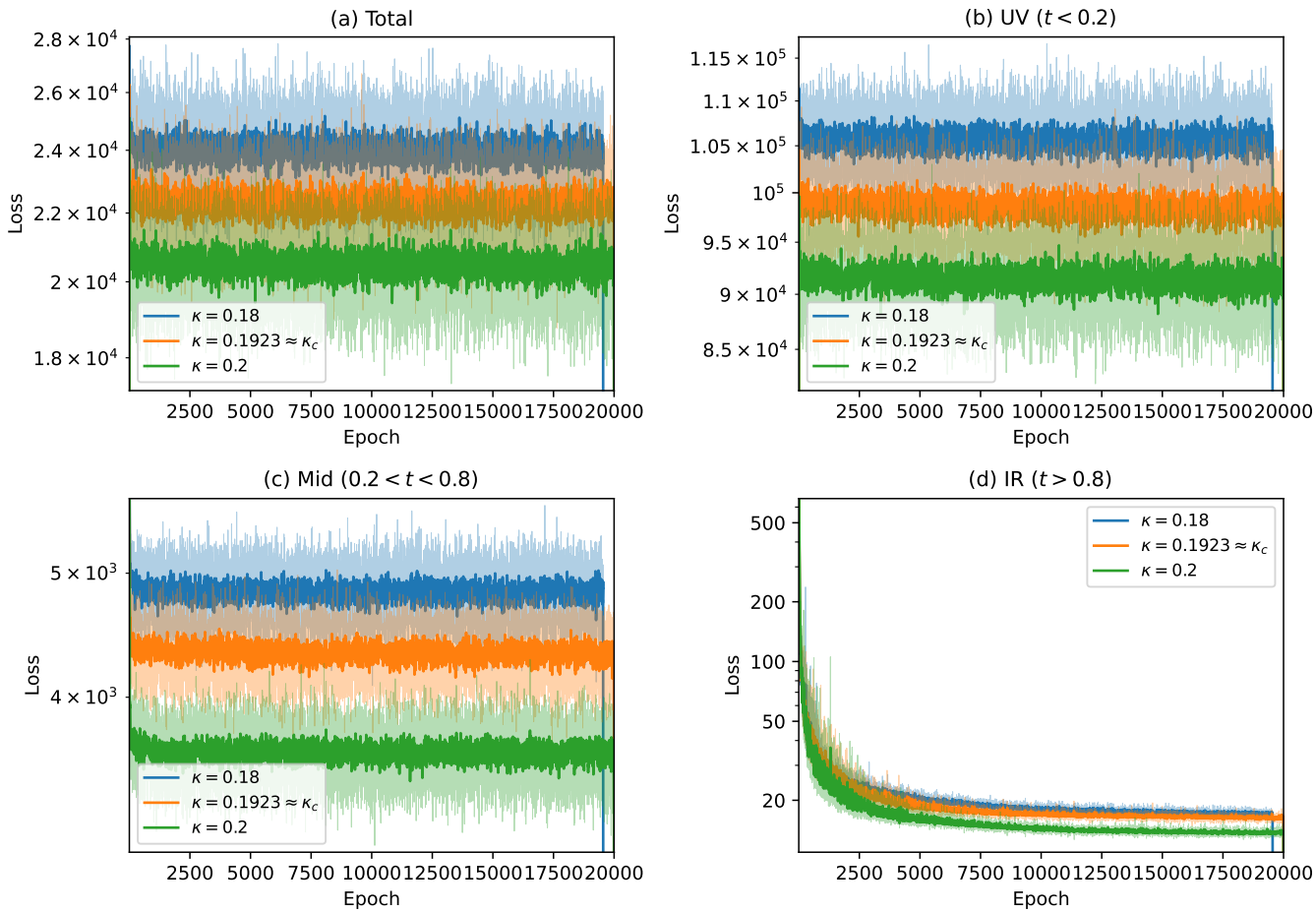

FIG. 14. Training loss vs. epoch for the 3D ϕ^4 NCSN++ diffusion model at $L = 64$, $\lambda = 0.9$: (a) total loss, (b) UV component ($t < 0.2$), (c) mid component ($0.2 < t < 0.8$), (d) IR component ($t > 0.8$). Each panel compares three κ values spanning the phase diagram.

point.

$$D_{\rm tot} = \sum_k D_k, \tag{32}$$

$$L_1^{\rm pw} = \sum_k k^{D-1} \big| G_{\rm DM}(k) - G_{\rm train}(k) \big|, \tag{33}$$

$$R_{\rm var} = \frac{\sum_k G_{\rm DM}(k)}{\sum_k G_{\rm train}(k)}. \tag{34}$$

$D_{\rm tot}$ is the Gaussian-surrogate KL of Eq. (29) summed over bins, and inherits its ratio-based scale invariance. $L_1^{\rm pw}$ is the L_1 norm of the phase-space-weighted density w_k of Eq. (31), and by Parseval's relation controls the absolute $\langle \phi^2 \rangle$ mismatch in physical units. $R_{\rm var}$ is the total-variance ratio, tracked directly rather than via $|R_{\rm var} - 1|$ because its signed deviation records whether the model underestimates or overestimates the bulk variance at each training stage. The three scalars carry complementary information. $D_{\rm tot}$ is insensitive to a uniform rescaling of G that $R_{\rm var}$ catches immediately. $L_1^{\rm pw}$ weights mid-k modes most strongly and is the closest of the three to a perceptual summary.

e. Perceptual benchmark: FID. For image data we cross-check the spectral diagnostics against the Fréchet Inception Distance (FID) [10]. Passing a sample set through the Inception-v3 network yields $d = 2048$-dimensional activations; fitting a Gaussian $\mathcal{N}(\mu, \Sigma)$ to these activations for both the DM and training samples, FID is the Fréchet distance between the two Gaussians

$$\mathrm{FID} = \|\mu_{\rm DM} - \mu_{\rm train}\|^2 + \mathrm{tr}\Big[\Sigma_{\rm DM} + \Sigma_{\rm train} - 2\big(\Sigma_{\rm DM}\Sigma_{\rm train}\big)^{1/2}\Big], \tag{35}$$

which, like $D_{\rm tot}$, is a Gaussian KL up to a monotone rescaling but in perceptual Inception-feature space rather than in momentum space. Comparing FID across training with the spectral scalars in Eqs. (32)eq:Rvar tests how much perceptual and physical-spectrum quality track each other.

f. Illustration on CelebA 64×64. Fig. 19 applies these three diagnostics to a NCSN++ diffusion model trained on CelebA grayscale 64×64 images, using log-spaced checkpoints from epoch 1 to epoch 10,000. Each checkpoint supplies $N = 512$ EM-sampled images; the reference $G_{\rm train}$ is computed from 50,000 training images,

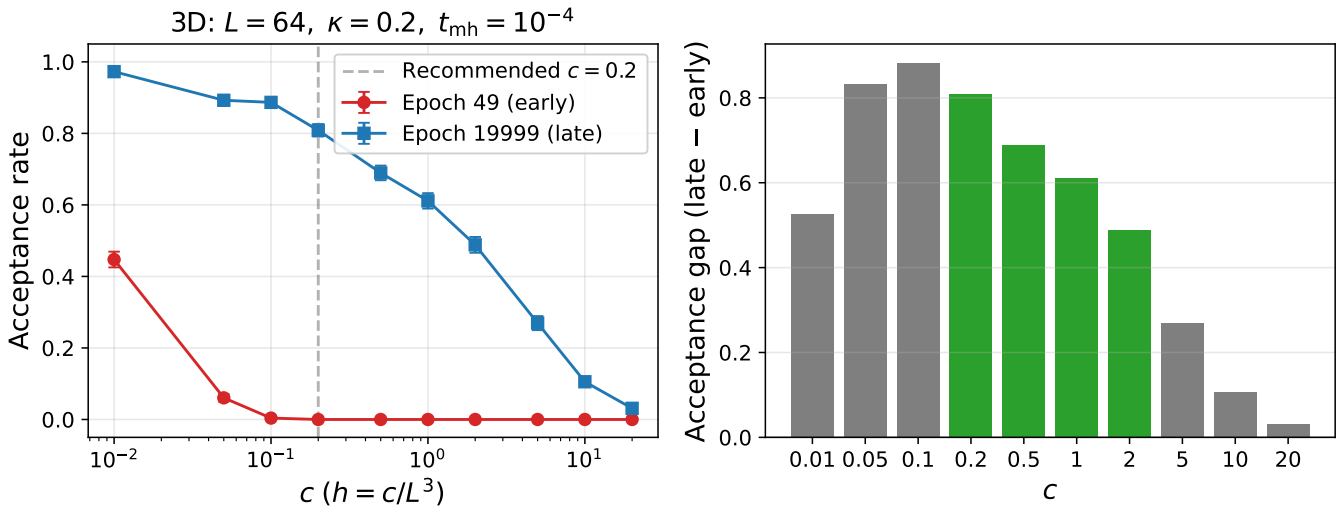

FIG. 15. Calibration of the MALA step size $h = c/L^3$ for the 3D ϕ^4 model at $L = 64$, $\kappa = 0.2$, $\lambda = 0.9$, with $t_{\rm mh} = 10^{-4}$, $N = 256$ and a single MH step. The optimal $c = 0.2$ matches the 2D result, validating the $h = c/L^D$ scaling.

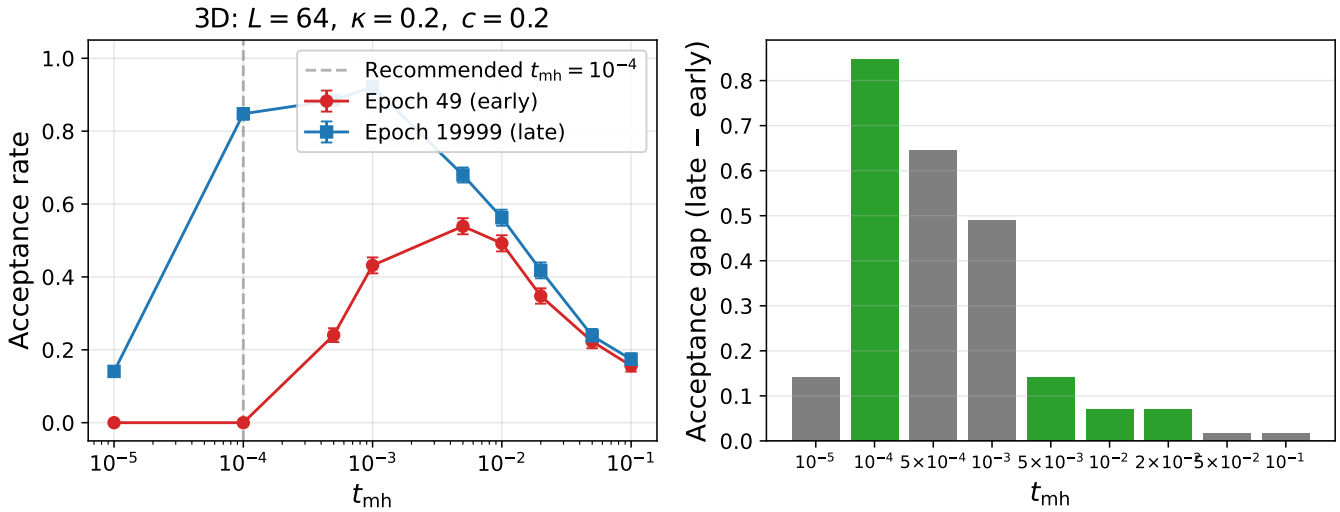

FIG. 16. Calibration of $t_{\rm mh}$ for the 3D model at $L = 64$, $\kappa = 0.2$, with $c = 0.5$ and $h = c/L^3$, $N = 256$ and a single MH step.

so $\sigma_{\rm train} \ll \sigma_{\rm DM}$ and the DM sample size dominates the bootstrap uncertainty. All three metrics display the expected monotonic improvement during training: the per-mode KL drops by nearly two orders of magnitude at all $|k|$, the z-score envelope contracts toward the $\pm 2\sigma$ band, and the phase-space weighted difference w_k shrinks and changes sign as the model passes from underestimating to slightly overestimating the low-k power. The UV (large-$|k|$) portion dominates the total KL throughout training, consistent with the difficulty of learning high-frequency image modes.

F. $\kappa = 0.26$: symmetric phase

Fig. 20 shows the propagator for $\kappa = 0.26$ at several training epochs compared to the HMC baseline. The diffusion model converges to the HMC propagator as training progresses, with the largest deviations observed at early epochs and at small momenta (large correlation lengths).

G. $\kappa = 0.2705$: near-critical regime

Near the critical point $\kappa \approx \kappa_c$, the correlation length diverges and the propagator develops a power-law tail $G(p) \sim p^{-(2-\eta)}$ at small momenta, making it the most

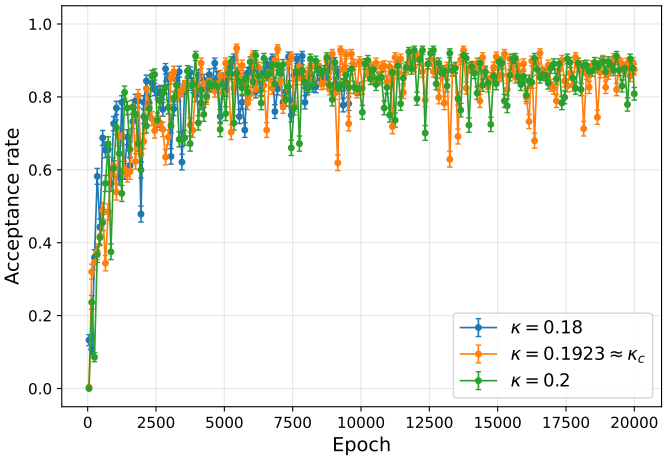

FIG. 17. MALA acceptance rate vs. training epoch for the 3D ϕ^4 model at $L = 64$, $\lambda = 0.9$, with $c = 0.5$, $h = c/L^3$, $t_{\rm mh} = 10^{-4}$, $N = 256$ and a single MH step.

demanding test case. Fig. 21 demonstrates that the diffusion model successfully captures this behavior, with the DM propagator approaching the HMC result upon sufficient training.

H. $\kappa = 0.28$: broken phase

In the ordered phase at $\kappa = 0.28$, the propagator acquires a large zero-mode contribution and a mass gap. Fig. 22 shows that the diffusion model also reproduces the propagator faithfully in this regime.

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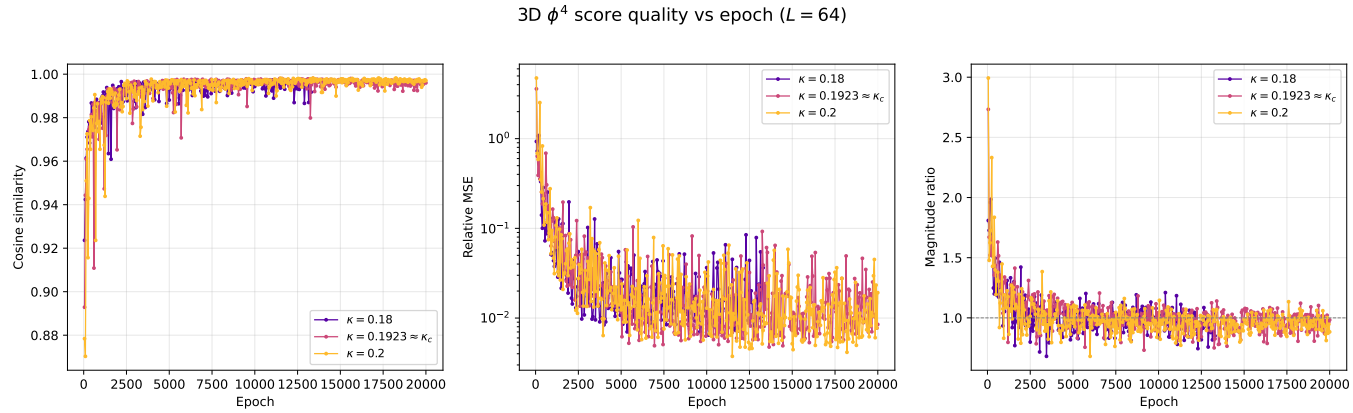

FIG. 18. Score quality vs. training epoch for the 3D ϕ^4 model at $L = 64$, $\lambda = 0.9$, evaluated at $t = 10^{-4}$ with $N = 64$ reference configurations. Left: cosine similarity. Centre: relative MSE. Right: magnitude ratio.

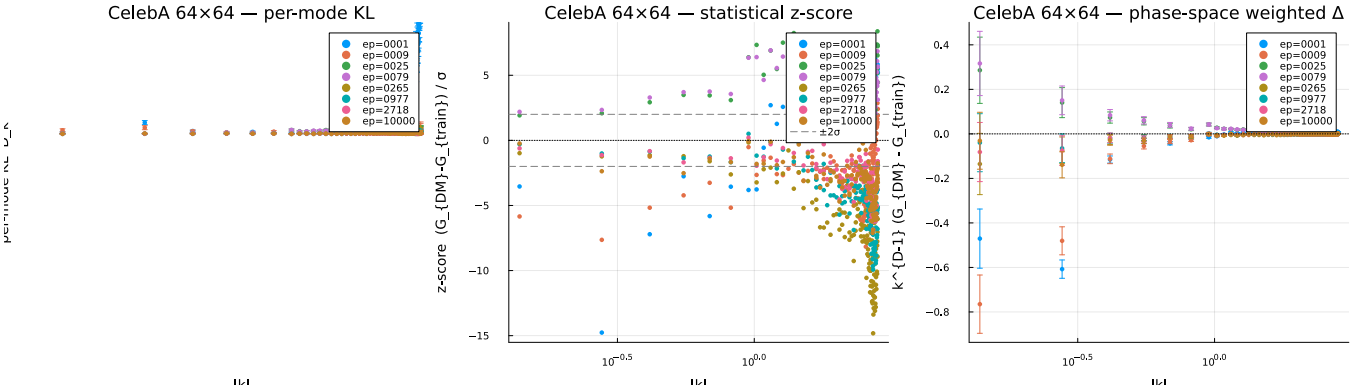

FIG. 19. Propagator-mismatch metrics for a NCSN++ diffusion model on CelebA grayscale 64×64 images at eight log-spaced training checkpoints. Left: per-mode Gaussian KL D_k defined in (29); the reference value $D_k = 0$ corresponds to exact propagator matching. Centre: statistical z-score of (30); the dashed lines mark the $\pm 2\sigma$ consistency band. Right: phase-space weighted propagator difference w_k of (31) with $D = 2$; integrating w_k over a momentum band returns the contribution of that band to the $\langle \phi^2 \rangle$ mismatch.

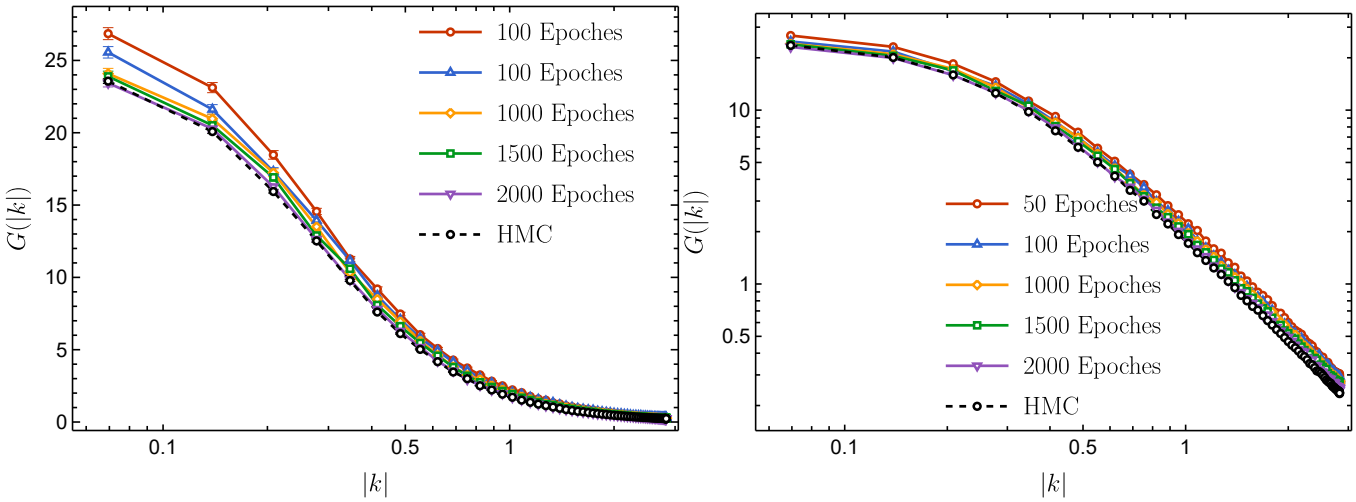

FIG. 20. Momentum-space propagator $G(p)$ for $\kappa = 0.26$ (symmetric phase), $L = 128$. Semi-log (left) and double-logarithmic (right) comparison of HMC and DM at different training epochs.

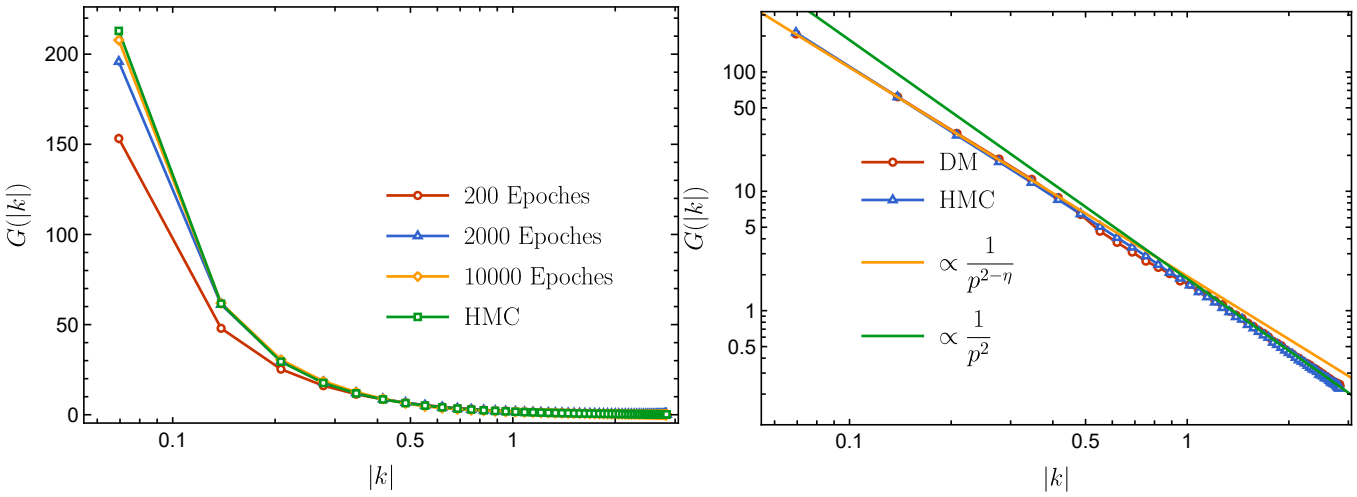

FIG. 21. Propagator $G(p)$ at $\kappa = 0.2705 \approx \kappa_c$ (near-critical), $L = 128$. Semi-log (left) and double-logarithmic (right) comparison of HMC and DM at different training epochs.

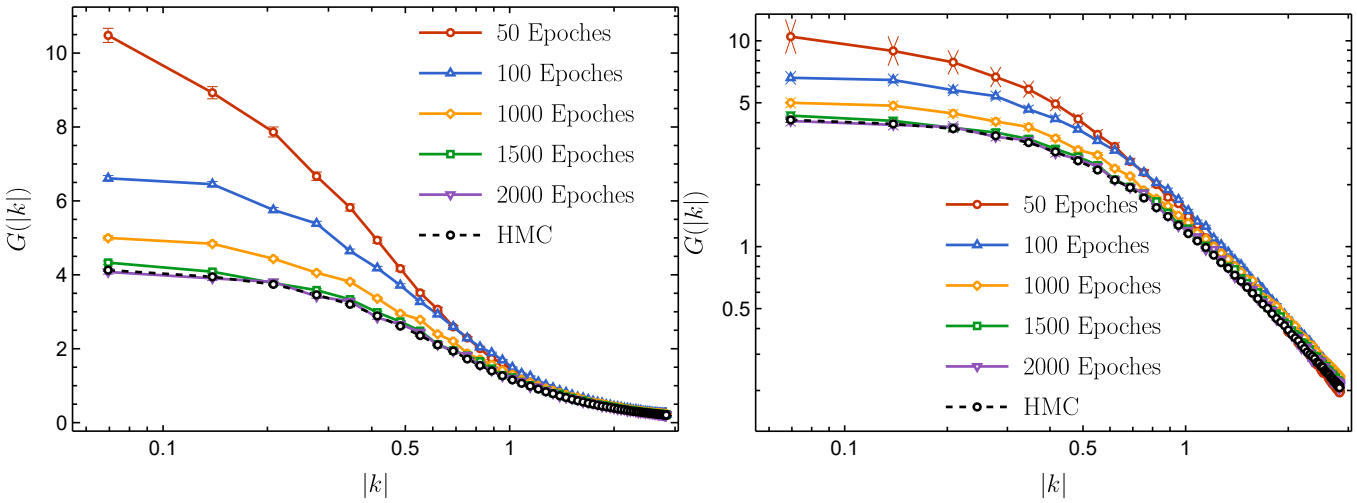

FIG. 22. Propagator $G(p)$ for $\kappa = 0.28$ (broken phase), $L = 128$. Semi-log (left) and double-logarithmic (right) comparison of HMC and DM at different training epochs.